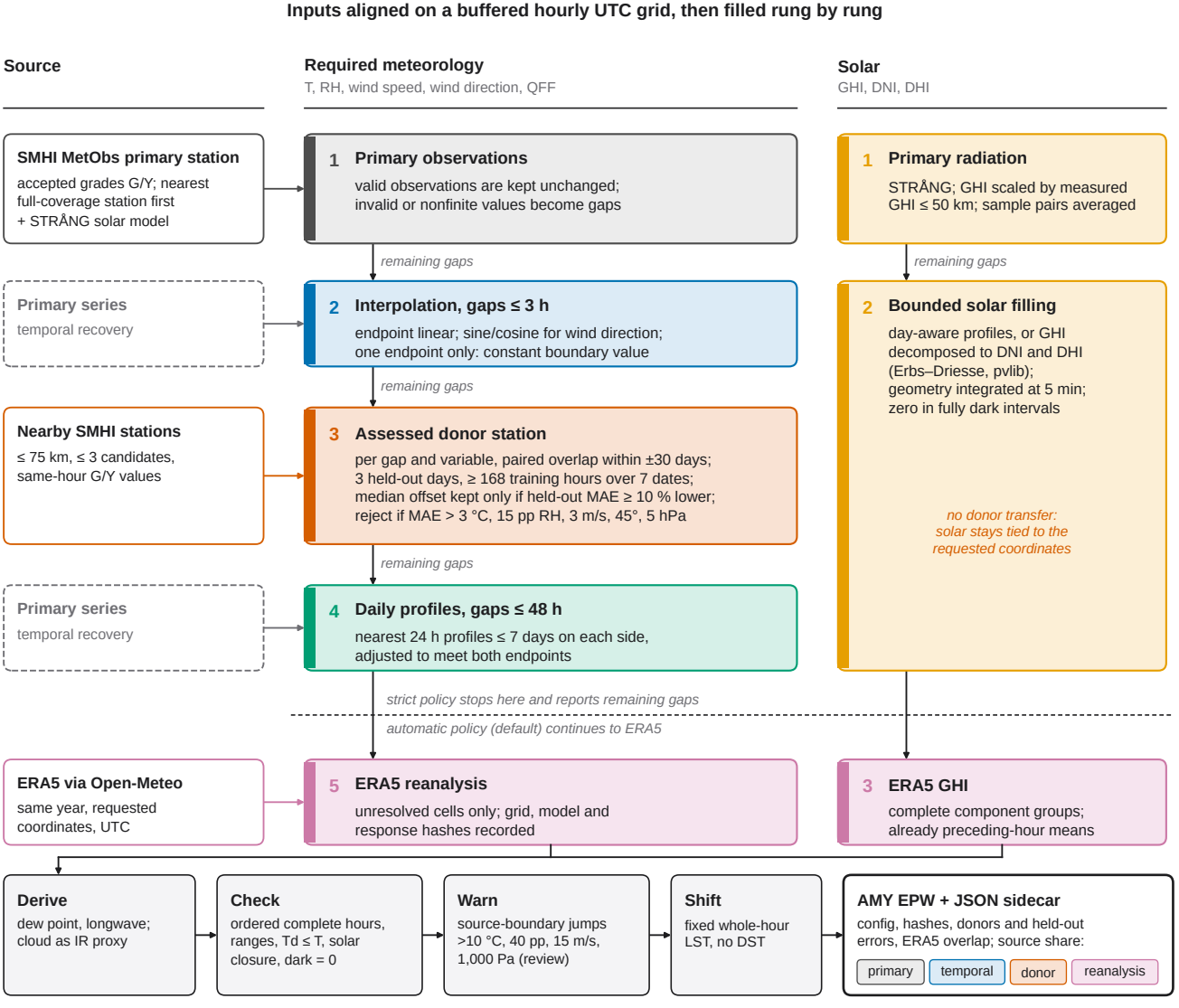
Inputs aligned on a buffered hourly UTC grid, then filled rung by rung
Source
Required meteorology
T, RH, wind speed, wind direction, QFF
Solar
GHI, DNI, DHI
SMHI MetObs primary station
accepted grades G/Y; nearest full-coverage station first + STRÅNG solar model
1 Primary observations
valid observations are kept unchanged; invalid or nonfinite values become gaps
1 Primary radiation
STRÅNG; GHI scaled by measured GHI ≤ 50 km; sample pairs averaged
remaining gaps
remaining gaps
Primary series
temporal recovery
2 Interpolation, gaps ≤ 3 h
endpoint linear; sine/cosine for wind direction; one endpoint only: constant boundary value
2 Bounded solar filling
day-aware profiles, or GHI decomposed to DNI and DHI (Erbs–Driesse, pvlib); geometry integrated at 5 min; zero in fully dark intervals
remaining gaps
Nearby SMHI stations
≤ 75 km, ≤ 3 candidates, same-hour G/Y values
3 Assessed donor station
per gap and variable, paired overlap within ±30 days; 3 held-out days, ≥ 168 training hours over 7 dates; median offset kept only if held-out MAE ≥ 10 % lower; reject if MAE > 3 °C, 15 pp RH, 3 m/s, 45°, 5 hPa
no donor transfer: solar stays tied to the requested coordinates
remaining gaps
Primary series
temporal recovery
4 Daily profiles, gaps ≤ 48 h
nearest 24 h profiles ≤ 7 days on each side, adjusted to meet both endpoints
strict policy stops here and reports remaining gaps
automatic policy (default) continues to ERA5
ERA5 via Open-Meteo
same year, requested coordinates, UTC
5 ERA5 reanalysis
unresolved cells only; grid, model and response hashes recorded
3 ERA5 GHI
complete component groups; already preceding-hour means
Derive
dew point, longwave; cloud as IR proxy
Check
ordered complete hours, ranges, Td ≤ T, solar closure, dark = 0
Warn
source-boundary jumps >10 °C, 40 pp, 15 m/s, 1,000 Pa (review)
Shift
fixed whole-hour LST, no DST
AMY EPW + JSON sidecar
config, hashes, donors and held-out errors, ERA5 overlap; source share:
primary
temporal
donor
reanalysis